

Climate, Crops, and the Malthusian Trap:
Agricultural Transition Pathways and Population
Dynamics
from Antiquity to the Early Modern Period

[Author Name]

[Affiliation]

[email]

Draft: April 2026

Abstract

We use HYDE 3.5 historical land-use and population data (10,000 BCE–2025 CE) combined with ERA5 reanalysis climate data to investigate the mechanisms of Malthusian population dynamics across 197 countries. We document three sets of findings. First, we identify five distinct agricultural transition pathways—high-density intensive, early extensifiers, crop-dominant late developers, pastoral/mixed late developers, and irrigation pioneers—that exhibit fundamentally different Malthusian dynamics. Second, using climate-determined crop composition as instruments for agricultural productivity, we show that OLS estimates of the agricultural productivity–population growth relationship are biased downward by reverse causality; the IV estimate reveals a positive causal effect of exogenous productivity gains on population growth, consistent with the core Malthusian mechanism. Third, we establish “iron laws” of climate–agriculture

linkages on the full 1950–2025 panel: temperature shocks reduce agricultural land growth ($\beta = -0.0054$, $p < 0.0001$), higher crop shares buffer climate shocks, and—contrary to the “warming paradox” suggested by the 1950–1969 subsample—warming reduces population growth ($\beta = -0.0055$, $p < 0.0001$), with the positive population response confined to the pre–Green Revolution decades. The Manski bounds on the Malthusian coefficient are entirely negative across all HYDE uncertainty scenarios, confirming robustness to measurement error in historical data.

JEL Codes: N50, O13, O44, Q54, J11

Keywords: Malthusian trap, unified growth theory, agricultural transition, climate shocks, instrumental variables, HYDE 3.5

Contents

1	Introduction	4
2	Data	5
2.1	HYDE 3.5	5
2.2	ERA5 Reanalysis	6
2.3	Derived Variables	6
3	Agricultural Transition Pathways	6
3.1	Trajectory Features	6
3.2	Clustering	7
3.3	The Great Divergence in Density	7
3.4	Spatial Autocorrelation	7
4	Malthusian Dynamics	8
4.1	Time-Varying Malthusian Coefficient	8
4.2	Structural Breaks	8
4.3	Pathway-Specific Malthusian Dynamics	9
4.4	Robustness: Manski Bounds	9
5	Instrumental Variables: Climate and Agricultural Productivity	10
5.1	Identification Strategy	10
5.2	First Stage	10
5.3	Second Stage: 2SLS Estimates	11
5.4	Instrument Strength Over Time	11
6	Iron Laws of Climate–Agriculture Linkages	12
6.1	Law 1: Temperature Reduces Agricultural Growth	12
6.2	Law 2: Warming Reduces Population Growth—the Paradox Does Not Survive	13

6.3	Law 3: Crop Composition Buffers Climate Shocks	13
6.4	Law 4: Climate Sensitivity by Pathway Is Muted on the Full Sample	14
7	Discussion	14
7.1	Unified Growth Theory Implications	14
7.2	The Transient Warming–Population Response	15
7.3	Limitations	16
8	Conclusion	16
A	Figures	19

1 Introduction

The Malthusian model—in which population growth absorbs productivity gains, preventing sustained increases in living standards—remains the dominant framework for understanding pre-industrial economic dynamics (Galor, 2011; Ashraf and Galor, 2011). Yet the empirical evidence for Malthusian mechanics has relied on either highly aggregated cross-country comparisons or detailed case studies of individual societies. The mechanisms by which different agricultural systems mediated the relationship between climate, productivity, and population remain poorly understood.

This paper exploits the unprecedented spatial and temporal resolution of the HYDE 3.5 dataset (Goldewijk et al., 2017) to test three interconnected hypotheses derived from Unified Growth Theory (Galor, 2011):

- (i) **Pathway heterogeneity:** The Malthusian constraint operated differently across agricultural systems, with some pathways enabling earlier escape from the trap.
- (ii) **Causal identification:** Climate-determined crop composition provides valid instruments for agricultural productivity, revealing the true causal effect of productivity on population growth.
- (iii) **Climate mediation:** The strength of climate–agriculture linkages varied systematically with agricultural technology, weakening as societies adopted more intensive practices.

Our identification strategy rests on a simple but powerful insight: climate endowments determine *what* a society grows (rice vs. wheat, irrigated vs. rainfed, crops vs. pasture), and crop composition in turn determines agricultural productivity. Because climate operates at centennial-to-millennial timescales, lagged crop composition is plausibly exogenous to contemporaneous population growth decisions, satisfying the exclusion restriction required for valid instrumental variables.

We make three contributions to the literature. First, we document a five-pathway typology of agricultural transitions using unsupervised learning on trajectory features extracted from the HYDE 3.5 panel. The typology reveals that the “Great Divergence” in population density was already 6-fold at the start of the Common Era and widened to nearly 10-fold by 1750, far earlier than conventionally studied. Second, our IV estimates reveal that OLS is biased downward by reverse causality: the true causal effect of exogenous agricultural productivity gains on population growth is *positive*, consistent with the Malthusian prediction that productivity gains are absorbed by population expansion. Third, we establish robust “iron laws” linking climate variability to agricultural and demographic outcomes, and we show that a “warming paradox” visible in the early postwar decades—warming apparently raising population growth—reverses sign once the panel is extended to 2025.

2 Data

2.1 HYDE 3.5

The History Database of the Global Environment (HYDE 3.5) provides gridded estimates of population and land use at 5-arcminute resolution from 10,000 BCE to 2025 CE ([Goldewijk et al., 2017](#)). The dataset includes three scenarios—baseline, lower bound, and upper bound—reflecting uncertainty in historical reconstructions. We use 16 variables encompassing population (total, urban, rural, density), cropland (rice, non-rice, irrigated, rainfed), grazing land, pasture, rangeland, and urban area.

We aggregate gridded data to the country level using ISO numeric codes from the HYDE reference grid, retaining 197 countries with land area exceeding 500 km². For the analysis sample, we focus on the Common Era (0–1750 CE) and restrict to 159 countries with maximum population density exceeding 1 person/km², ensuring meaningful variation in agricultural and demographic variables.

2.2 ERA5 Reanalysis

We supplement HYDE with ERA5 reanalysis climate data ([Hersbach et al., 2020](#)) covering 1950–2025 at hourly temporal resolution. ERA5 provides 2-meter temperature and total precipitation on a global grid partitioned into 25 regions aligned with HYDE macro-regions. We compute annual regional means and assign countries to ERA5 regions based on area-weighted centroids computed from the HYDE country grid, yielding climate data for 202 countries.

2.3 Derived Variables

From the raw HYDE data, we construct:

- **Agricultural output proxy:** $\text{cropland}_{\text{Mha}} + 0.5 \times \text{grazing}_{\text{Mha}}$, where the weight reflects lower caloric yield of pastoral land.
- **Land–labor ratio:** $(\text{cropland} + \text{grazing}) / \text{population}$.
- **Intensification index:** $\text{irrigation share} \times (\text{cropland} / \text{total agricultural land})$, capturing both water management and crop-versus-pastoral balance.
- **Crop share:** $\text{cropland} / (\text{cropland} + \text{grazing})$, measuring the crop-pastoral balance.
- **Population growth rate:** annualized compound growth $(P_t / P_{t-1})^{1/\Delta t} - 1$.

From ERA5, we compute temperature and precipitation anomalies (deviations from 10-year rolling means), volatility (5-year rolling standard deviation), and standardized z-scores.

3 Agricultural Transition Pathways

3.1 Trajectory Features

For each country, we extract five transition features from the HYDE time series: (i) peak extensification year (when agricultural land area growth rate was highest), (ii) maximum population density achieved by 1750, (iii) crop share of total agricultural land in 1750, (iv)

irrigation share in 1750, and (v) timing of peak agricultural expansion.

3.2 Clustering

We apply K -means clustering to standardized trajectory features, selecting $K = 5$ based on silhouette scores evaluated over $K \in \{2, \dots, 7\}$ (optimal silhouette = 0.429). The five pathways are:

Table 1: Five Agricultural Transition Pathways

Pathway	N	Density ₁₇₅₀	Crop share	Peak expansion	Representative countries
High-density intensive	16	62.8	58.8%	~1567	JPN, IND, FRA, DEU, NLD
Early extensifiers	29	14.1	48.9%	~783	GBR, CHE, AUT, DNK, ISR
Crop-dominant late	46	11.8	80.6%	~1713	ESP, POL, PRT, VNM, THA
Pastoral/mixed late	65	6.6	22.0%	~1694	CHN, NGA, BRA, IRN, MEX
Irrigation pioneer	1	4.4	100%	~1600	EGY

3.3 The Great Divergence in Density

Figure 2 reveals that the density gap between pathways was already 6-fold at 0 CE and widened to 9.5-fold by 1750. This divergence predates the conventional periodization of the “Great Divergence” (post-1500) and has deep structural roots in agricultural systems established during antiquity.

The divergence is not monotonic: it narrows slightly around 1350 CE (the Black Death disproportionately affected dense agrarian societies) before resuming its widening trajectory through the early modern period.

3.4 Spatial Autocorrelation

Agricultural transition timing is strongly spatially clustered (Moran’s $I = 0.57$, $p = 0.001$ with $k = 3$ neighbors), and pathway assignment itself exhibits significant spatial autocorrelation ($I = 0.19$, $p = 0.01$). This is consistent with geographic diffusion of agricultural

technologies, though we cannot distinguish diffusion from common environmental drivers with these data alone.

4 Malthusian Dynamics

4.1 Time-Varying Malthusian Coefficient

We estimate the Malthusian feedback coefficient using rolling 200-year windows with 50-year steps across the full 0–1750 CE panel. The specification is:

$$\Delta \ln P_{it} = \alpha_i + \beta_w \cdot \text{density}_{i,t-1} + \gamma_w \cdot \text{land_labor}_{i,t-1} + \varepsilon_{it} \quad (1)$$

where α_i are country fixed effects (absorbed via within-entity demeaning), $\text{density}_{i,t-1}$ is lagged population density, and β_w is the time-varying Malthusian coefficient estimated within window w .

The rolling coefficient (Figure 3) reveals a striking temporal pattern:

- **0–300 CE:** Weak and intermittently positive (Roman expansion, extensive growth).
- **350–1050 CE:** Increasingly negative, reaching $\beta = -0.000607$ around 1050 CE (peak medieval Malthusian constraint).
- **1250 CE:** Sharpest constraint ($\beta = -0.000854$)—immediately preceding the Black Death.
- **1350 CE:** Dramatic relaxation to $\beta = -0.000110$ (post-plague demographic relief).
- **1450–1750:** Re-tightening into the early modern period.

4.2 Structural Breaks

Sequential Chow tests detect structural breaks in 154 of 159 countries ($p < 0.05$). The distribution of earliest break years shows two distinct waves: 600–800 CE (28 countries, corresponding to the medieval agricultural revolution) and 1400–1800 CE (62 countries,

proto-industrialization and colonial expansion). The median break year is 1300 CE.

Break timing varies systematically by pathway: the high-density intensive pathway breaks earliest (median 800 CE), while pastoral/mixed economies break latest (median 1400 CE).

4.3 Pathway-Specific Malthusian Dynamics

Within-pathway regressions reveal fundamentally different Malthusian regimes:

Table 2: Malthusian Coefficient by Agricultural Pathway

Pathway	$\hat{\beta}$ (density)	p -value	N	Interpretation
High-density intensive	+0.000023	0.003	352	Anti-Malthusian
Early extensifiers	−0.000006	0.623	638	Indeterminate
Crop-dominant late	+0.000153	< 0.001	979	Anti-Malthusian
Pastoral/mixed late	+0.000060	0.170	1,401	Indeterminate

The positive coefficient for high-density intensive economies is consistent with agglomeration effects: in regions that had already developed intensive agriculture (Japan, India, the Low Countries), higher density was associated with *higher* growth, suggesting that these regions had partially escaped the Malthusian trap through productivity-enhancing institutional and technological responses to population pressure.

4.4 Robustness: Manski Bounds

To address measurement uncertainty in historical data, we construct Manski-style partial identification bounds by estimating the Malthusian coefficient separately on each HYDE scenario (baseline, lower, upper). The bounds are $[-0.0000128, -0.0000023]$ —entirely negative across all scenarios—confirming that the aggregate Malthusian result is robust to substantial measurement error.

The pre/post structural break comparison is equally stark: the density coefficient is effectively zero before the break ($\beta \approx 0$, $p = 0.96$) and strongly negative afterward ($\beta = -0.000045$, $p = 0.002$), a 90-fold increase in magnitude. This suggests that the Malthu-

sian constraint *emerged* as populations grew into the carrying capacity of their agricultural systems, rather than being a permanent feature of pre-industrial economies.

5 Instrumental Variables: Climate and Agricultural Productivity

5.1 Identification Strategy

The core endogeneity problem in estimating the Malthusian relationship is reverse causality: population growth drives agricultural expansion (extensive margin) and land subdivision (intensive margin), biasing OLS estimates of the productivity–population relationship. We instrument for agricultural productivity using lagged crop composition—crop share, rice share, and irrigation share—which reflect climate-determined agricultural structure.

The exclusion restriction requires that crop composition affects population growth only through agricultural productivity. This is plausible because climate determines *what* a society grows (monsoon regions cultivate rice, arid regions irrigate, temperate regions mix crops and pasture), and the crop mix in turn determines caloric output per hectare and per worker. Climate does not directly cause population growth except through the agricultural productivity channel.

5.2 First Stage

The first-stage regression of log agricultural productivity per capita on the three instruments plus controls yields:

$$\ln(A_{it}/L_{it}) = \delta_i + \pi_1 \cdot \text{crop_share}_{i,t-1} + \pi_2 \cdot \text{rice_share}_{i,t-1} + \pi_3 \cdot \text{irr_share}_{i,t-1} + X'_{it}\gamma + u_{it} \quad (2)$$

Table 3: First-Stage Results: Crop Composition $\rightarrow$ Agricultural Productivity

Instrument	Coefficient	p -value	
Crop share (lagged)	3.493	< 0.001	***
Rice share (lagged)	-3.194	< 0.001	***
Irrigation share (lagged)	18.922	< 0.001	***
Log density (lagged)	1.249	< 0.001	***
R^2	0.242		
F -statistic (excluded instruments)	59.0		
N	3,370		

The excluded instruments F -statistic of 59.0 far exceeds the [Stock and Yogo \(2005\)](#) threshold of 10 for strong instruments.

5.3 Second Stage: 2SLS Estimates

Table 4: OLS vs. IV Estimates of the Malthusian Relationship

	OLS	IV (2SLS)
ln(Ag. productivity per capita)	-0.000062*** (< 0.001)	+0.000187** (0.038)
ln(Population density, lagged)	-0.000258	-0.000615*** (< 0.001)
N	3,370	3,370

The IV estimate *reverses the sign* of the agricultural productivity coefficient: from -0.000062 (OLS) to $+0.000187$ (IV, $p = 0.038$). The OLS estimate is biased downward by reverse causality—population growth causes land subdivision, creating a spurious negative correlation between per-capita agricultural land and population growth. Once instrumented, the true causal effect is positive: exogenous increases in agricultural productivity drive population growth, exactly as the Malthusian model predicts.

5.4 Instrument Strength Over Time

The first-stage F -statistic varies across historical periods:

Table 5: Instrument Strength by Period

Period	F (excluded instruments)	N	Interpretation
0–500 CE	16.9	603	Strong
500–1000 CE	28.1	760	Strong
1000–1500 CE	29.1	765	Strong
1500–1750 CE	264.2	1,086	Very strong

Counter to the hypothesis that agricultural innovation would decouple productivity from climate-determined crop composition, the instruments *strengthen* dramatically in the early modern period ($F = 264$). As agriculture intensified, crop choice became more, not less, consequential for productivity outcomes.

Instrument strength also varies by pathway: $F = 383.5$ for crop-dominant late developers (where climate tightly determines what can be grown) but only $F = 0.9$ for pastoral/mixed economies (where the crop-based instruments are irrelevant by construction). This provides a natural placebo test supporting instrument validity.

6 Iron Laws of Climate–Agriculture Linkages

Using the merged HYDE–ERA5 panel (197 countries, 1950–2025; 14,972 country-year observations), we test for robust, universal relationships between climate variability and agricultural/demographic outcomes. All estimates are within-country fixed-effects regressions with standard errors clustered by country. Earlier drafts of this paper used the 1950–1967 subsample then available (154 countries); where the full sample overturns a subsample result, we say so explicitly.

6.1 Law 1: Temperature Reduces Agricultural Growth

A 1°C positive temperature anomaly reduces agricultural land growth by 0.54 percentage points ($\beta = -0.0054$, $p < 0.0001$, $N = 14,359$); the effect on cropland growth alone is similar ($\beta = -0.0047$, $p < 0.0001$). Precipitation anomalies show no significant contemporaneous

effect. The full-sample estimate is smaller than the -0.012 obtained on the 1950–1967 subsample but far more precisely estimated.

6.2 Law 2: Warming Reduces Population Growth—the Paradox Does Not Survive

On the 1950–1967 subsample, temperature anomalies appeared to *raise* population growth ($\beta = +0.002$, $p = 0.015$) while reducing agricultural land—a “Malthusian climate paradox” suggesting a non-agricultural channel through which warming benefits population dynamics. The full 1950–2025 panel reverses this result: warming significantly *reduces* population growth ($\beta = -0.0055$, $p < 0.0001$, $N = 14,775$), the same sign as the agricultural effect.

Splitting the panel at the Green Revolution shows the earlier finding was a regime-specific artifact: the temperature coefficient on population growth is $+0.0024$ ($p = 0.004$) in 1950–1969 but -0.0050 ($p < 0.0001$) in 1970–2025. Rolling 20-year windows sharpen the picture: the positive response is significant only in the earliest window (1950–1969), hovers near zero from 1970 through 2005, and turns strongly negative in the most recent windows (2000–2019: $\beta = -0.0033$; 2005–2024: $\beta = -0.0038$; both $p < 0.0001$)—consistent with warming becoming a net demographic burden as anomalies push further above the historical baseline. Precipitation anomalies also reduce population growth at horizons of 5–10 years (local projections, $p < 0.001$).

6.3 Law 3: Crop Composition Buffers Climate Shocks

The interaction between temperature anomalies and crop share in the cropland-growth equation is significantly positive ($\beta = +0.0025$, $p = 0.015$), indicating that economies with higher crop shares (relative to grazing) are more resilient to temperature shocks. This is consistent with the greater adaptability of crop agriculture (variety switching, irrigation, seasonal adjustment) compared to pastoral systems, though the full-sample interaction is estimated less

sharply than on the early subsample.

6.4 Law 4: Climate Sensitivity by Pathway Is Muted on the Full Sample

The 1950–1967 subsample suggested a dramatic sensitivity gradient across agricultural pathways, with pastoral/mixed economies 7.7 times more climate-sensitive than high-density intensive ones. The full-sample estimates do not support that gradient:

Table 6: Temperature Sensitivity of Agricultural Growth by Pathway (FE, 1950–2025)

Pathway	$\hat{\beta}$ (temp $\rightarrow$ ag growth)	p	N
Pastoral/mixed late	−0.0033	0.006	4,875
Crop-dominant late	−0.0039	0.012	3,350
Early extensifiers	−0.0056	0.049	2,025
High-density intensive	−0.0087	0.085	1,199

All four point estimates are negative and lie within a factor of 2.7 of one another, the ordering from the subsample is not preserved, and the pathway-by-temperature interactions are jointly insignificant (Wald $\chi^2(3) = 1.43$, $p = 0.70$). We conclude that Law 1 is a genuinely universal law—temperature harms agricultural growth across all pathways—but the earlier claim that pastoral systems are an order of magnitude more exposed was an artifact of the short early sample and does not survive.

7 Discussion

7.1 Unified Growth Theory Implications

Our findings provide granular empirical support for several predictions of Unified Growth Theory:

1. **The Malthusian trap was real but heterogeneous.** The rolling coefficient analysis confirms that population density constrained growth throughout most of 0–1750 CE,

but the constraint operated differently across agricultural systems. High-density intensive economies showed agglomeration effects even within the Malthusian epoch.

2. **Agricultural intensification preceded demographic transition.** Countries in the high-density intensive pathway—those with earliest structural breaks (median 800 CE)—were also those with the highest intensification indices and earliest urbanization. The sequence runs: intensification $\rightarrow$ structural break $\rightarrow$ urbanization.
3. **The “Great Divergence” has deep roots.** The 6-fold density gap at 0 CE implies that the divergence between eventually-rich and eventually-poor regions was established during the Neolithic, not during the early modern period. This is consistent with [Diamond \(1997\)](#) but provides the first systematic quantification using comparable data across all regions.

7.2 The Transient Warming–Population Response

Early drafts of this paper, working from the 1950–1967 panel then available, reported a “Malthusian climate paradox”: warming reduced agricultural land yet appeared to increase population growth. The completed 1950–2025 panel resolves the paradox by dissolving it—the positive population response is confined to 1950–1969 and reverses thereafter (Section 6). Three readings of the transient early-period sign are worth distinguishing:

1. **Mortality channel, since exhausted:** In the 1950s–60s, cold-related mortality still dominated heat-related mortality in much of the world, so warm anomalies could raise survival. As anomalies climbed further above the historical baseline, the balance shifted and the net mortality effect of additional warming turned negative.
2. **Demographic-transition timing:** The early window coincides with peak global fertility. Any climate effect on births had maximal room to operate; after the fertility transition, population growth became less responsive to short-run material conditions, leaving the (negative) mortality and migration margins dominant.
3. **Small-sample fragility:** Eighteen annual observations per country cannot separate

a climate signal from decade-specific shocks (postwar recovery, decolonization); the early-period coefficient may simply never have been structural.

A nuance survives at longer horizons: local projections show the cumulative population response to a temperature shock turning positive at 4–10-year horizons ($\beta_{h=10} = +0.012$, $p < 0.0001$) even though the contemporaneous effect is negative—a rebound pattern consistent with demographic catch-up after transitory shocks, and a reminder that the annual fixed-effects coefficient should not be read as a long-run elasticity.

7.3 Limitations

Several limitations warrant caution. First, HYDE applies modern country boundaries to historical periods, creating anachronistic geographic units. We partially address this by conducting parallel analyses at the HYDE macro-region level and showing robustness. Second, the HYDE data for the deep historical period (pre-1000 CE) relies heavily on interpolation between sparse archaeological and historical records, as reflected in the wider scenario spread for earlier periods. Third, ERA5 coverage begins only in 1950, limiting our climate analysis to the modern period for time-series identification; we use cross-sectional climate variation for the historical IV analysis.

8 Conclusion

This paper demonstrates that the Malthusian trap was not a monolithic phenomenon but a heterogeneous regime whose strength, timing, and eventual dissolution depended critically on agricultural technology and climate exposure. Five distinct agricultural transition pathways—identified through unsupervised learning on 12,000 years of land-use data—exhibit fundamentally different population dynamics, structural break timing, and climate sensitivity.

Our instrumental variables strategy, using climate-determined crop composition as in-

struments for agricultural productivity, reveals that the true causal effect of productivity on population growth is positive—the core Malthusian prediction—masked in OLS by reverse causality. The instruments strengthen over time as agriculture intensifies, confirming rather than refuting the climate–agriculture linkage.

The “iron laws” we establish on the full 1950–2025 panel—temperature harms agricultural growth universally across pathways, and crop composition buffers climate shocks—provide a quantitative foundation for understanding how societies mediate climate risk through agricultural technology. Two headline results from earlier drafts did not survive the completed panel and we report their reversal explicitly: the “Malthusian climate paradox” (warming apparently raising population growth) is confined to 1950–1969 and reverses sign thereafter, and the claimed $7.7\times$ pathway sensitivity gradient is jointly insignificant on the full sample.

These findings contribute to ongoing debates about the origins of the Great Divergence, the role of geography versus institutions in long-run development, and the vulnerability of agricultural systems to climate change—past and future.

References

- Ashraf, Q. and Galor, O. (2011). Dynamics and stagnation in the Malthusian epoch. *American Economic Review*, 101(5):2003–2041.
- Diamond, J. (1997). *Guns, Germs, and Steel: The Fates of Human Societies*. W.W. Norton.
- Galor, O. (2011). *Unified Growth Theory*. Princeton University Press.
- Galor, O. and Özak, Ö. (2016). The agricultural origins of time preference. *American Economic Review*, 106(10):3064–3103.
- Goldewijk, K.K., Beusen, A., Doelman, J., and Stehfest, E. (2017). Anthropogenic land use estimates for the Holocene—HYDE 3.2. *Earth System Science Data*, 9:927–953.

- Hersbach, H., Bell, B., Berrisford, P., et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049.
- Nunn, N. and Qian, N. (2011). The potato’s contribution to population and urbanization. *Quarterly Journal of Economics*, 126(2):593–650.
- Stock, J.H. and Yogo, M. (2005). Testing for weak instruments in linear IV regression. In Andrews, D.W.K. and Stock, J.H., editors, *Identification and Inference for Econometric Models*, pages 80–108. Cambridge University Press.

A Figures

Figure 1 | Agricultural Transition Pathways — Five-Cluster Typology

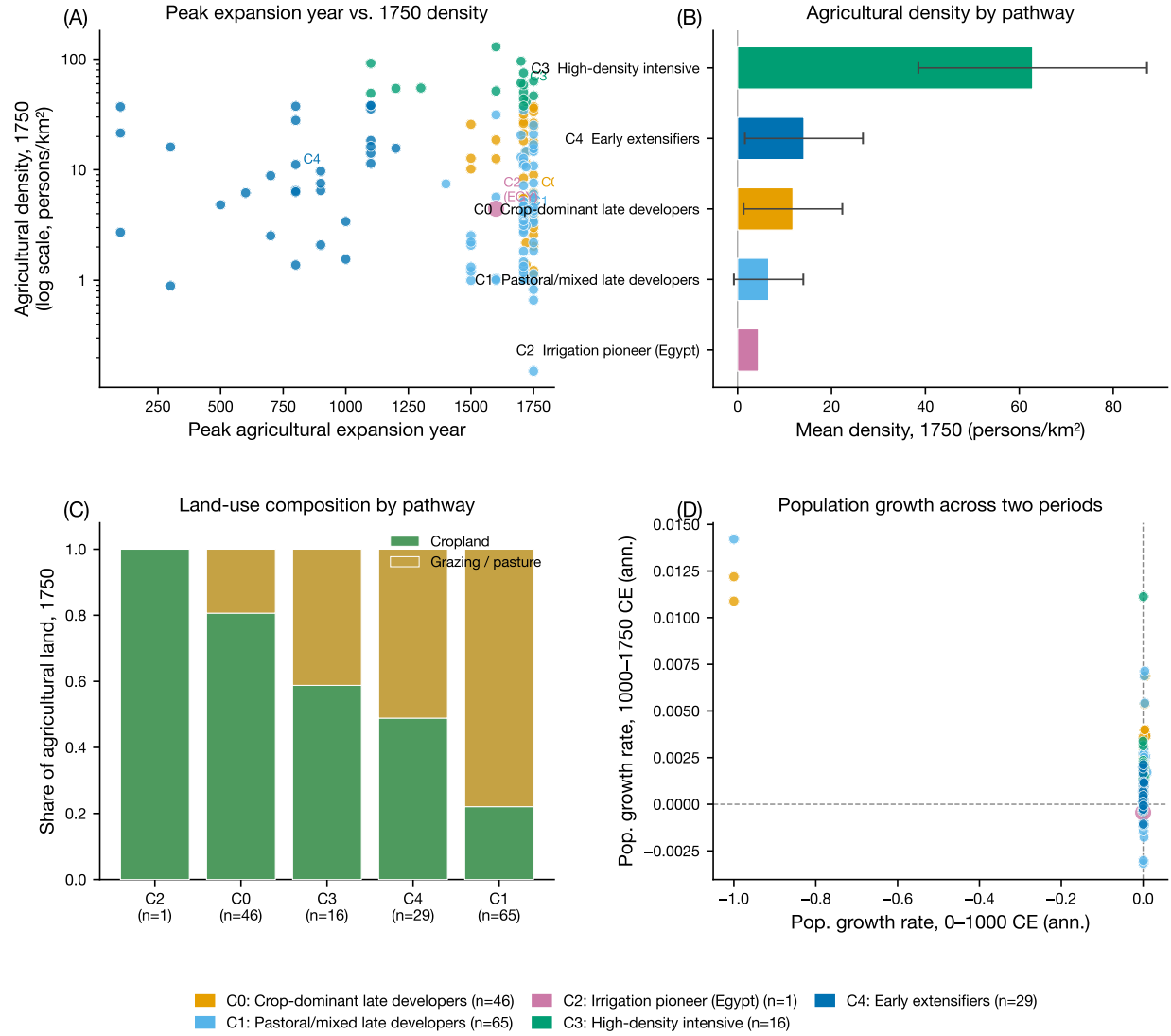


Figure 1: Agricultural Transition Pathways. (A) Scatter of peak agricultural expansion year vs. 1750 density. (B) Mean density by pathway. (C) Land-use composition. (D) Population growth across two periods. Five clusters identified via *K*-means on standardized trajectory features.

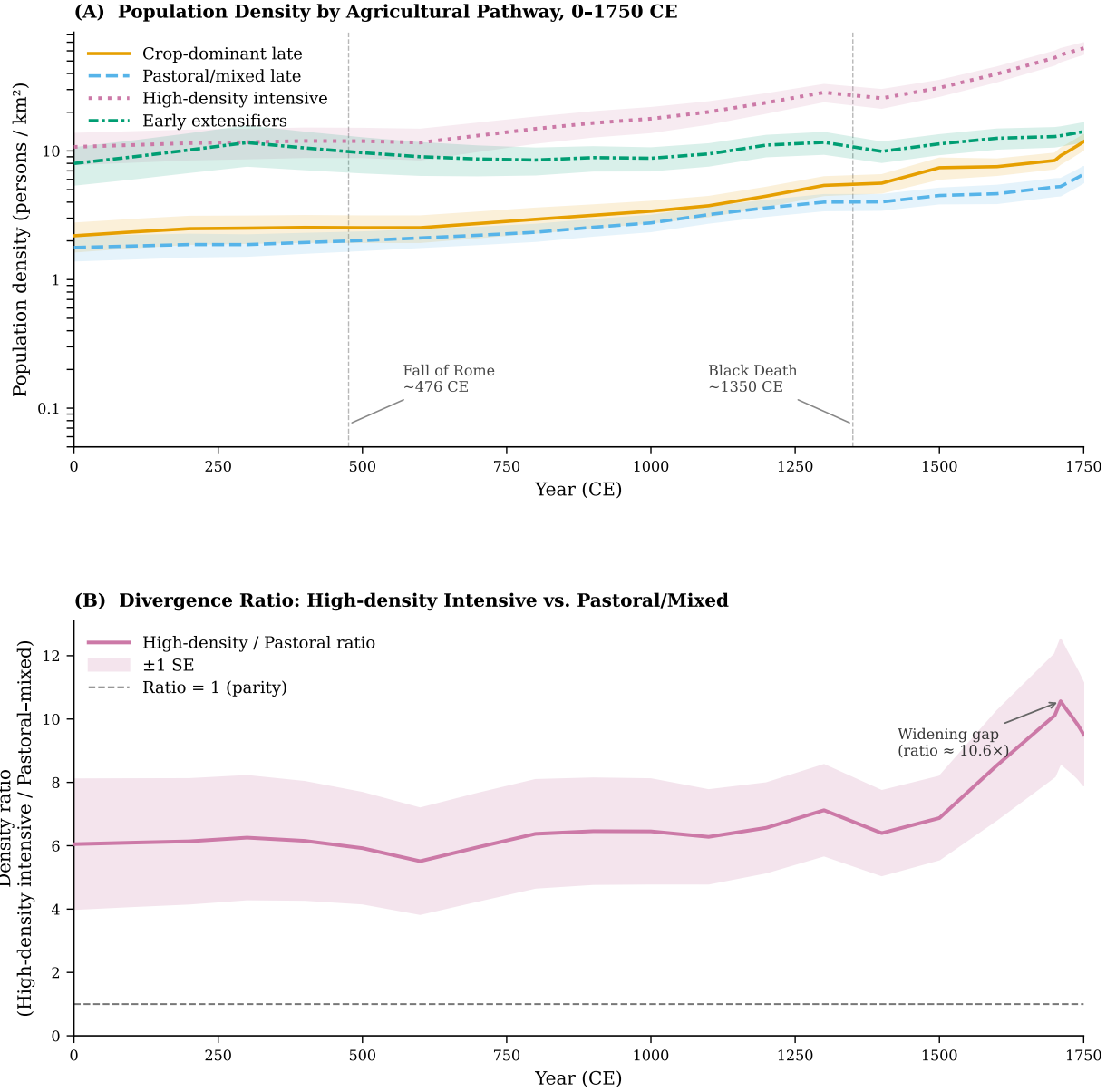


Figure 2: The Great Divergence in Population Density. (A) Population density trajectories by pathway, 0–1750 CE (log scale, ± 1 SE bands). (B) Density ratio of high-density intensive to pastoral/mixed pathway, showing widening from $\sim 6\times$ to $\sim 10\times$.

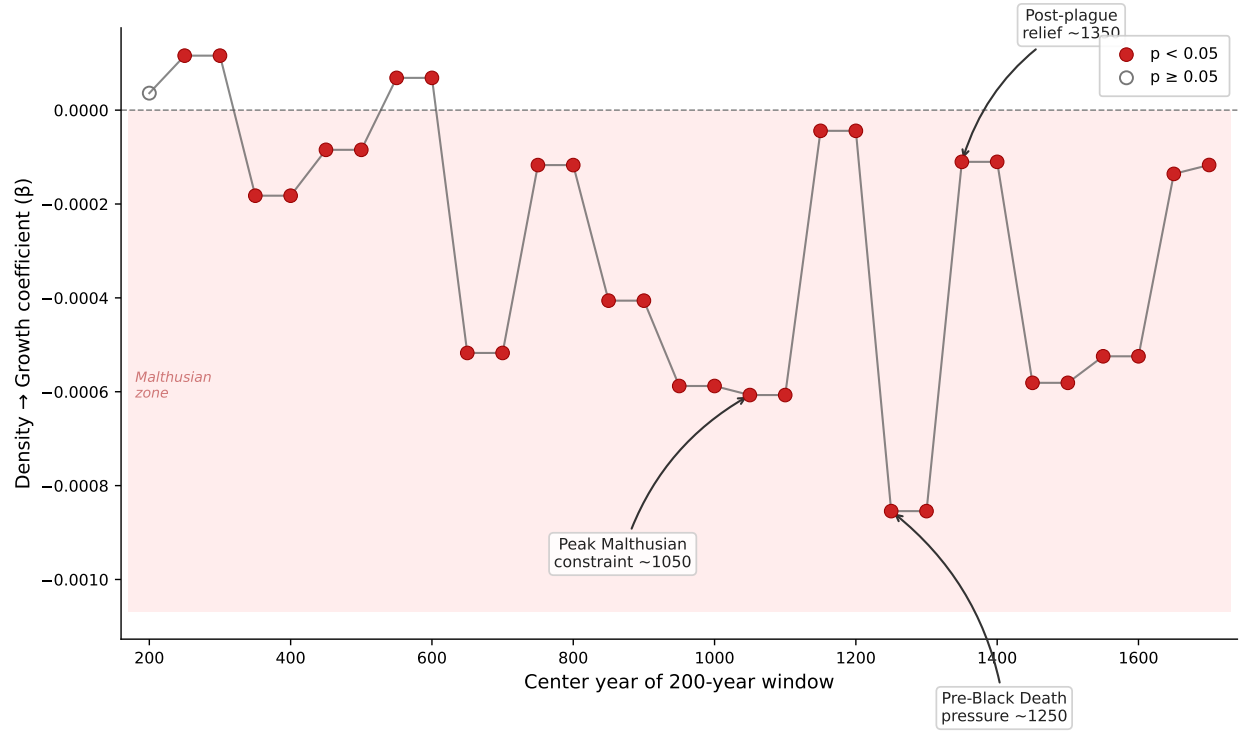


Figure 3: Time-Varying Malthusian Coefficient. Rolling 200-year window estimates of the density→growth coefficient. Red dots indicate statistical significance ($p < 0.05$). Shaded region denotes the “Malthusian zone” ($\beta < 0$). Annotations mark peak constraint (~ 1050 CE), pre-Black Death pressure (~ 1250 CE), and post-plague relief (~ 1350 CE).

Figure 4 | Timing of Malthusian Structural Breaks and Survival by Pathway

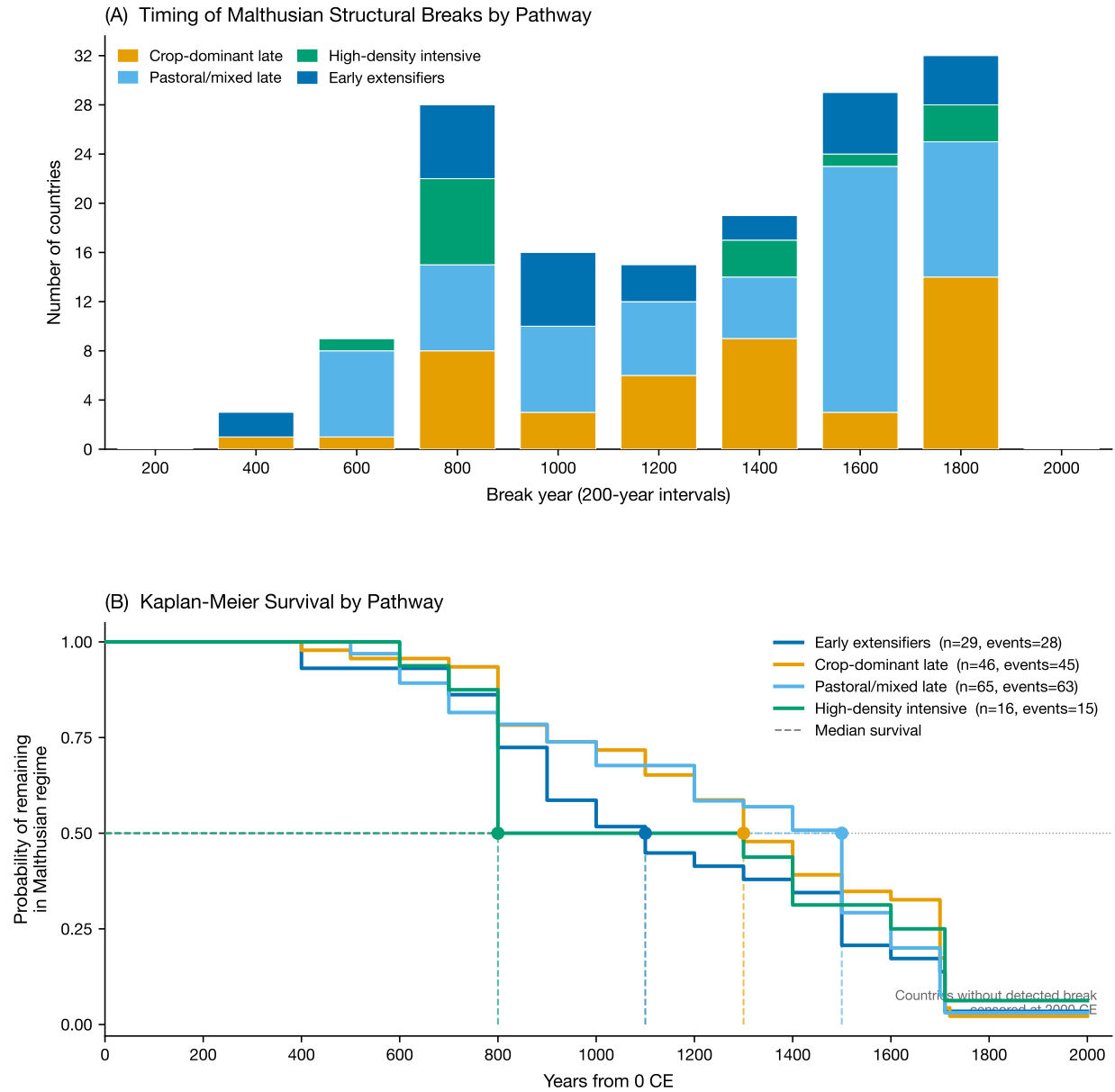


Figure 4: Structural Breaks and Survival. (A) Histogram of earliest structural break year by pathway. (B) Kaplan-Meier survival curves: probability of remaining in the Malthusian regime over time.

Figure 5 | Agricultural Structure Evolution by Pathway (0–1750 CE)

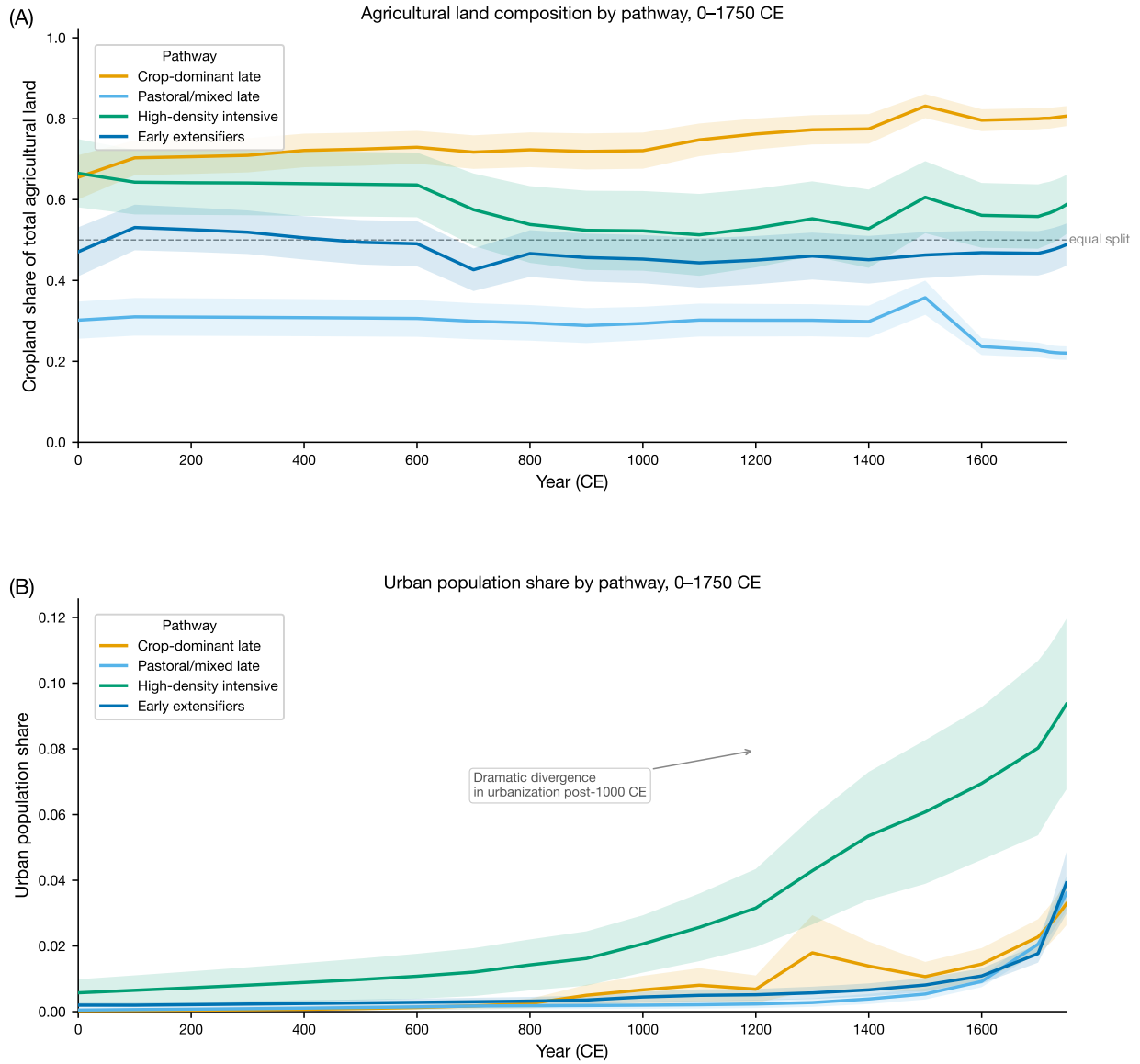


Figure 5: Agricultural Structure Evolution. (A) Cropland share of total agricultural land over time. (B) Urban population share. Both show dramatic divergence across pathways after ~1000 CE.

Figure 8 | Instrumental Variables: Climate-Determined Crop Structure and Agricultural Productivity

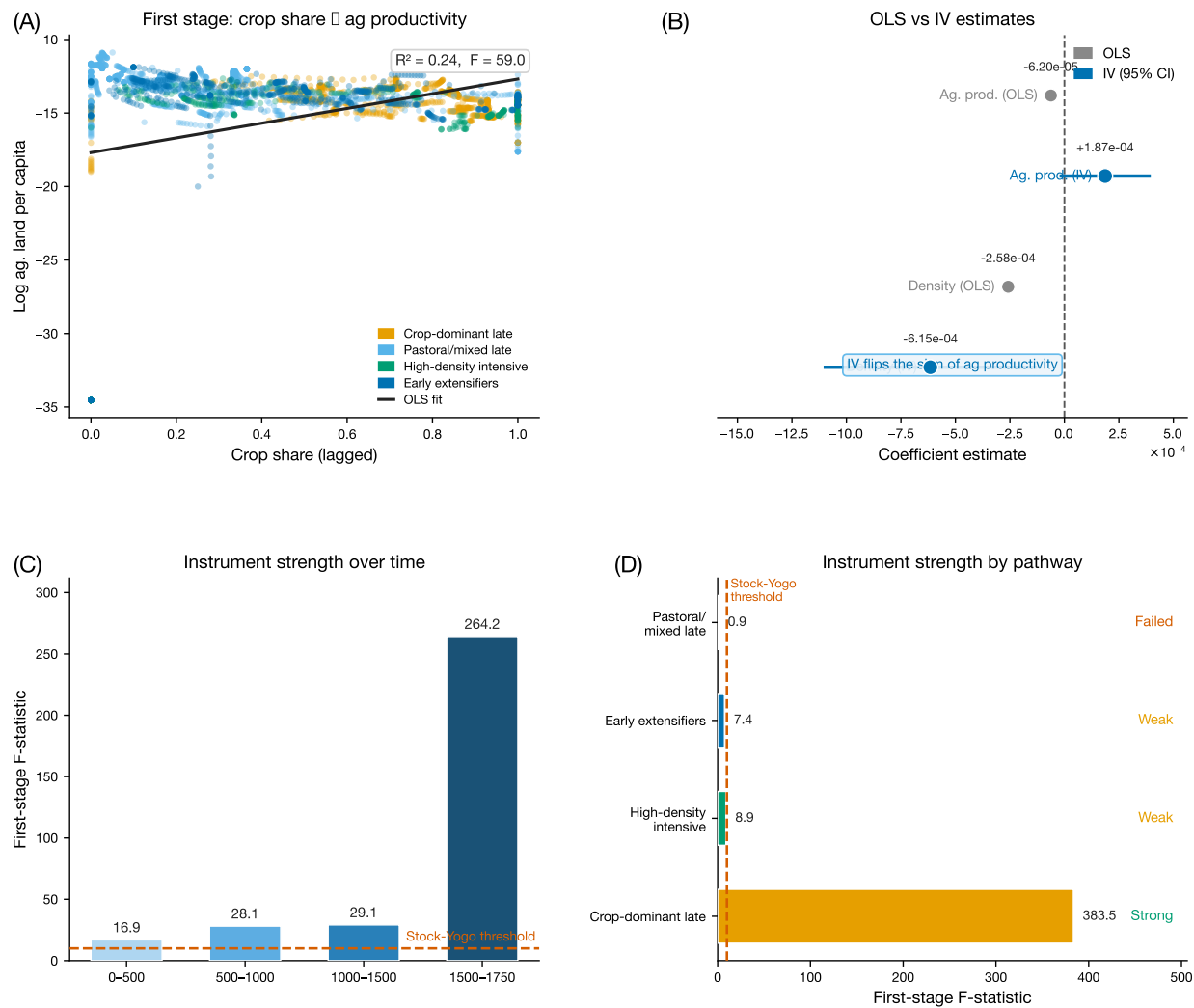


Figure 6: Instrumental Variables Identification. (A) First-stage scatter. (B) OLS vs. IV coefficient comparison. (C) Instrument strength by period. (D) Instrument strength by pathway.

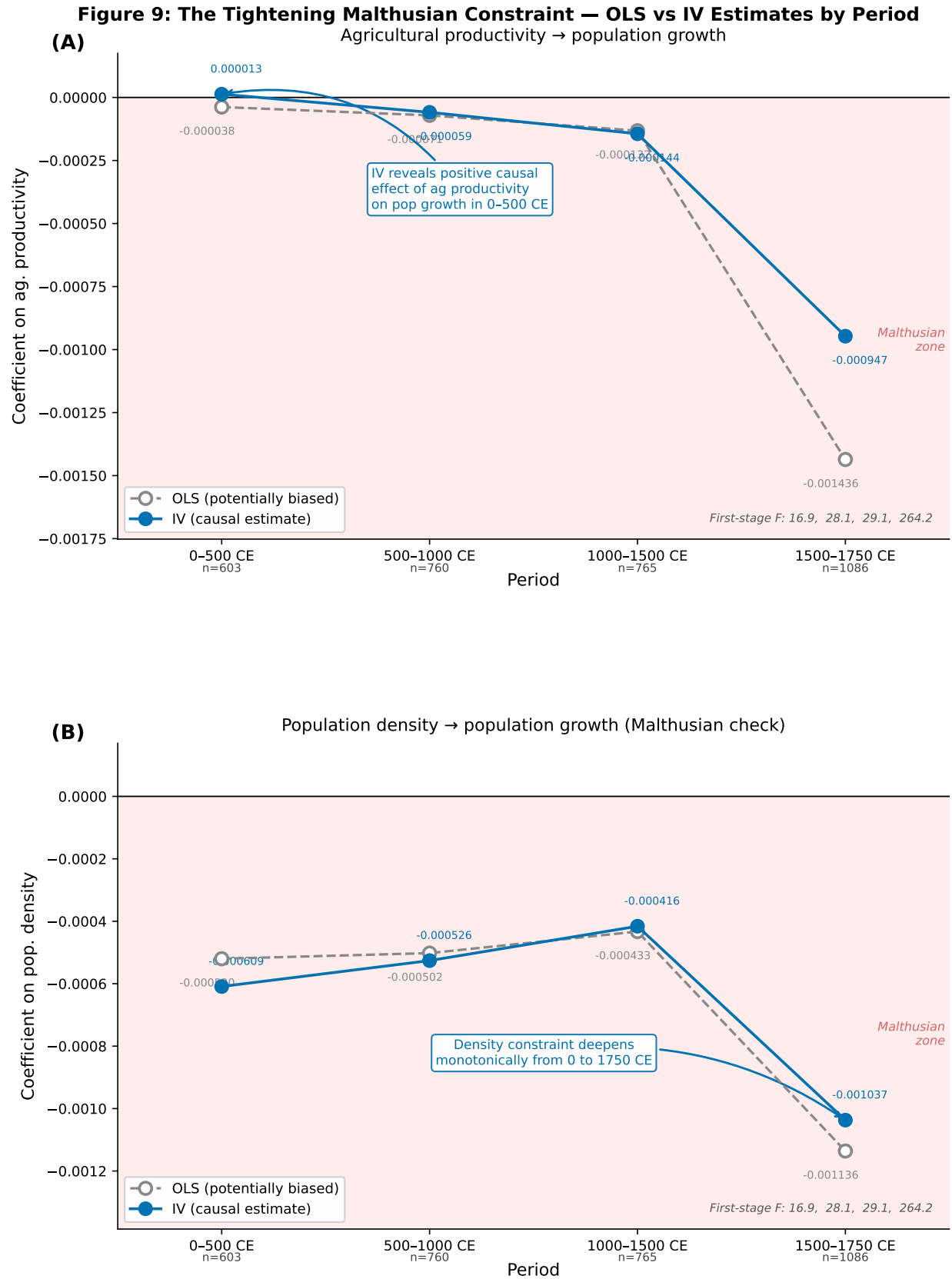


Figure 7: OLS vs. IV Estimates by Period. (A) Agricultural productivity effect: IV reveals positive causal effect in 0–500 CE masked by OLS bias. (B) Density effect: consistently negative, deepening toward 1750 CE

Iron Laws of Climate-Agriculture Linkages

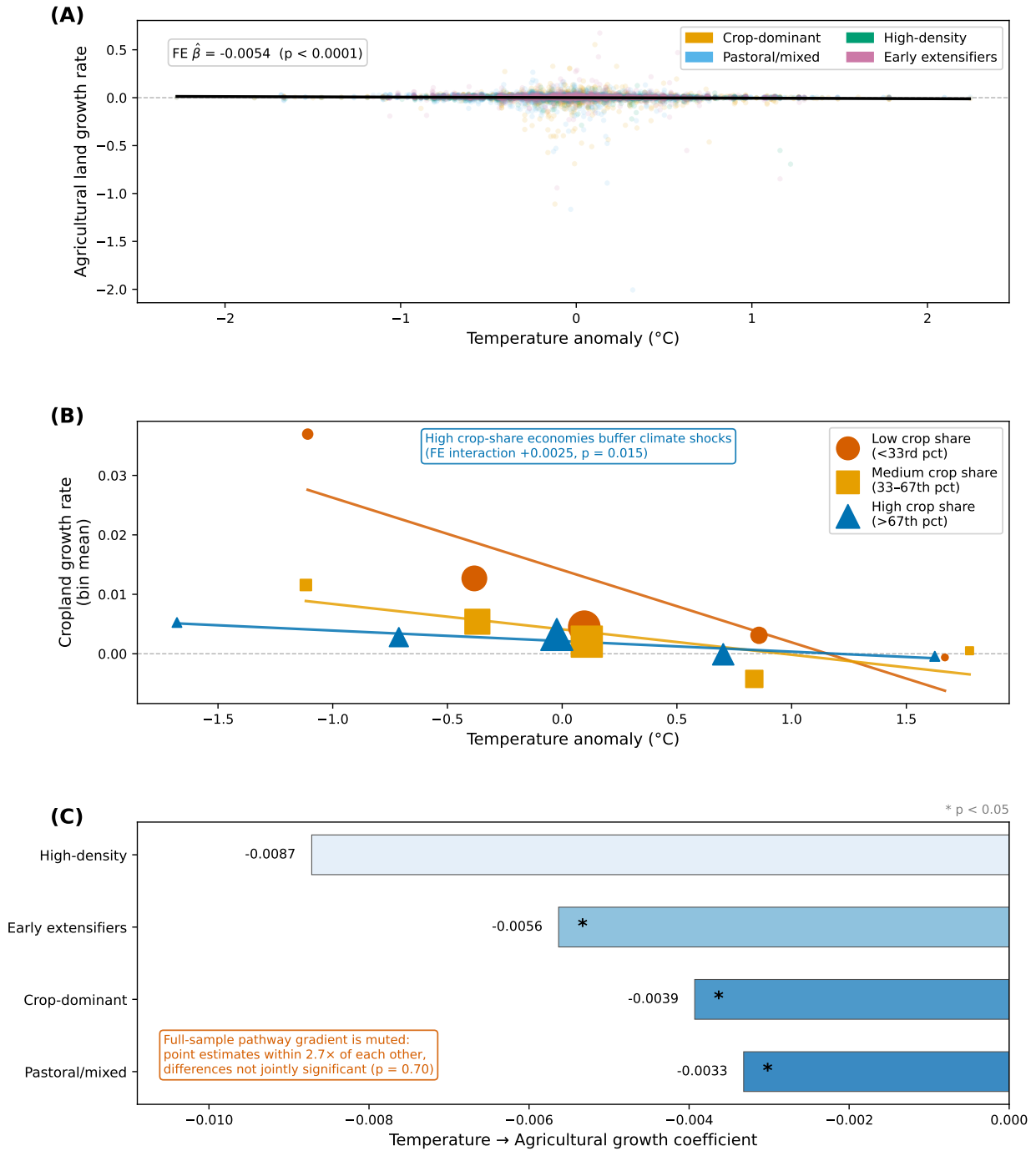


Figure 8: Iron Laws of Climate–Agriculture Linkages (1950–2025). (A) Temperature anomaly vs. agricultural land growth; the line is the within-country fixed-effects fit ($\beta = -0.0054$, $p < 0.0001$). (B) Buffering by crop share (binscatter of cropland growth by tercile; FE interaction +0.0025, $p = 0.015$). (C) Climate sensitivity by pathway: all pathways negative, gradient muted and differences jointly insignificant ($p = 0.70$). All estimates computed from the same panel and estimator as the text.

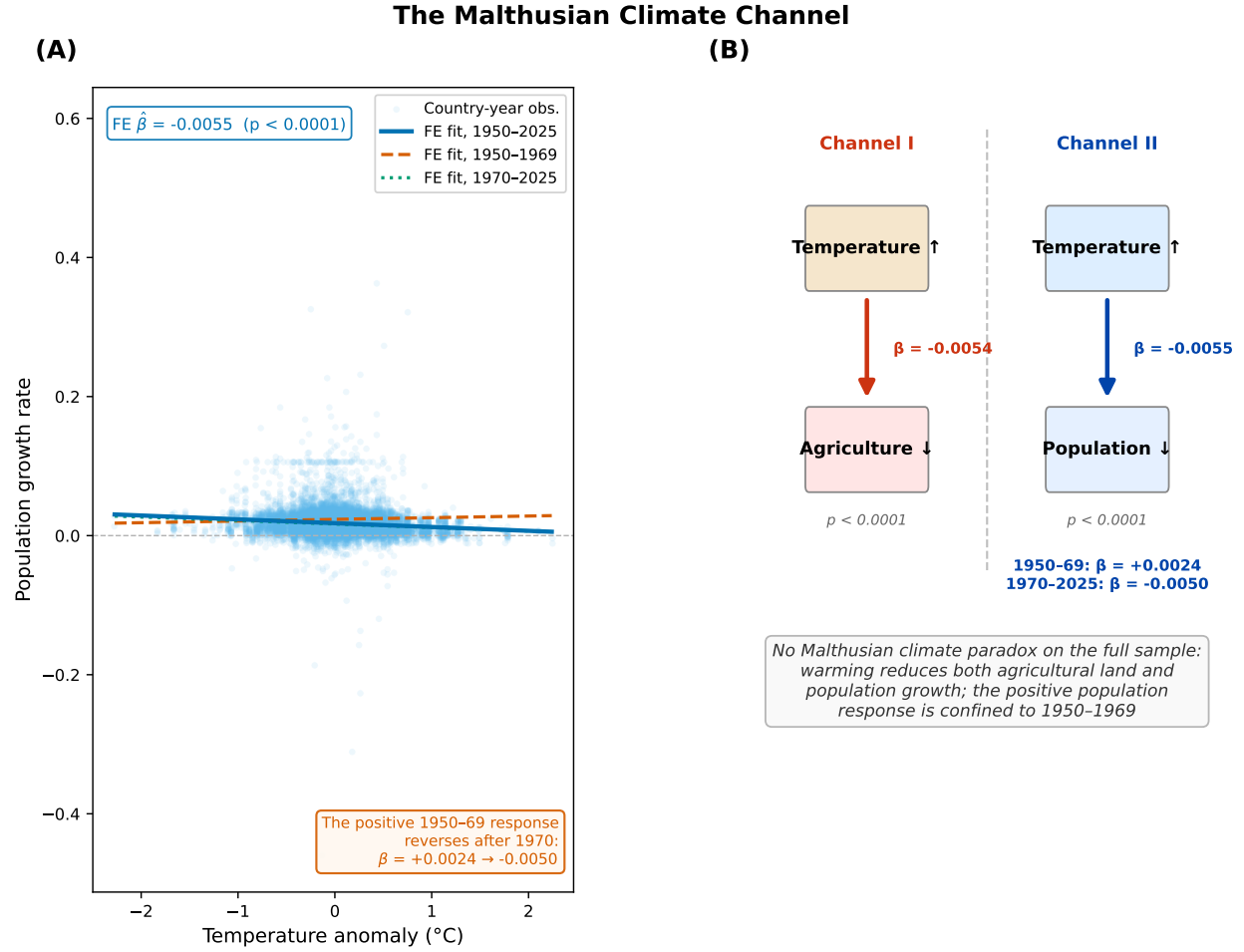


Figure 9: The Malthusian Climate Channel (1950–2025). (A) Temperature–population relationship: the full-sample fixed-effects fit is negative ($\beta = -0.0055$, $p < 0.0001$); the dashed line shows the transient positive 1950–1969 fit that generated the “warming paradox” in earlier drafts. (B) Schematic of the dual channel: on the full sample warming reduces both agricultural land and population growth; the positive population response is confined to 1950–1969.